MODEL DEVELOPMENT / ARTIFACT 01

Algorithm types, connected.

10 algorithms and model families across tabular data, vision, language, and generative AI.

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READ THE FRAMEWORK: learning style + domain + task

Learning style describes the training objective, not a permanent property of an architecture. Broad supervised/unsupervised categories appear alongside the specific training setup. In self-supervised training, targets come from the data itself. Conditional models and later fine-tuning can add supervision. Domain labels show selected common uses, not every possible application. All use cases are illustrative, not reported experiments.

01 Linear Regression [1]

Supervised

Tabular

Fits a weighted linear relationship between input features and a numeric target.

Use: Estimate a home price from area, location features, and property characteristics.

02 Random Forest [1]

Supervised

Tabular

Trains randomized decision trees on resampled data, then averages predictions or votes.

Use: Predict customer churn from account activity and service history.

03 Gradient Boosting [1]

Supervised

Tabular

Adds weak learners sequentially, each improving the current ensemble against a chosen loss.

Use: Rank loan applications by default risk using labeled historical outcomes.

04 Convolutional Neural Network [2]

Supervised: classifier setup

Computer Vision

Learns shared local filters and progressively combines visual features to predict a class.

Use: Flag visible manufacturing defects in labeled product photographs.

05 K-Means [1]

Unsupervised

Tabular / Computer Vision

Alternates assigning points to the closest centroid and updating each cluster mean.

Use: Group customers by scaled purchasing behavior without predefined segment labels.

06 Principal Component Analysis [1]

Unsupervised

Tabular / Computer Vision

Projects centered data onto orthogonal directions that retain the most variance.

Use: Reduce correlated sensor measurements to fewer features for visualization.

07 BERT [3]

Unsupervised/self-supervised pretraining; supervised fine-tuning

NLP

Uses bidirectional context to recover masked tokens; original BERT also used a sentence-pair objective.

Use: Fine-tune a pretrained encoder on labeled support tickets to classify intent.

08 GPT-style Transformer [4]

Unsupervised data / self-supervised pretraining

NLP / Generative AI

Uses causal attention to predict the next token from preceding tokens, then generates sequentially.

Use: Draft a product description from a supplied brief, followed by human review.

09 Generative Adversarial Network [5]

Unsupervised: unconditional setup

Computer Vision / Generative AI

A generator synthesizes samples while a discriminator learns to distinguish generated from real data.

Use: Generate synthetic product-image variations for creative exploration.

10 Diffusion Model [6]

Unsupervised data / self-supervised denoising

Computer Vision / Generative AI

Learns to reverse added noise; repeated denoising transforms a random sample into generated content.

Use: Generate image concepts; text-conditioned variants guide generation with prompts.

CHOOSE BY THE TASK

Predict a known outcome: supervised models. Discover groups: K-Means. Compress features: PCA. Understand text: BERT-based classification. Generate new content: GPT, GANs, or diffusion. Evaluate predictive models on held-out data; assess generated outputs for quality, diversity, and factual reliability.

Source key: [1] scikit-learn User Guide [2] He et al., 2015 [3] Devlin et al., 2019 [4] Radford et al., 2019 [5] Goodfellow et al., 2014 [6] Ho et al., 2020. Context: [7] Costa & Trojahan, 2024. Full references and selection notes accompany this infographic.